

ZAG

Avatar Generation App

Architecture · Wireframes · Prompt System

STACK

Next.js · Eleven v3 · Kling 2.0

CHARACTER

ZAG — Black Crocodile Mascot

BRAND

Must Be Nuts

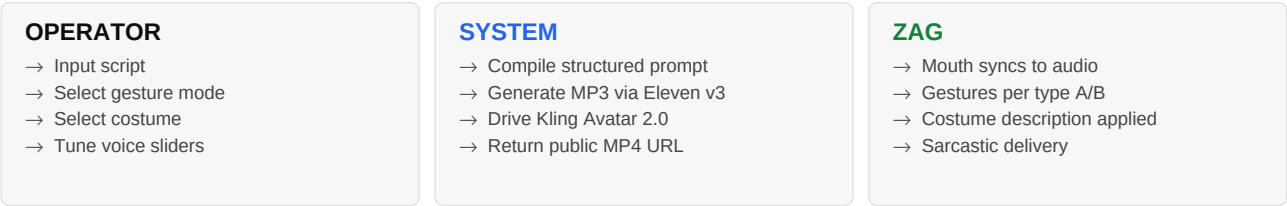
OUTPUT

9:16 vertical MP4

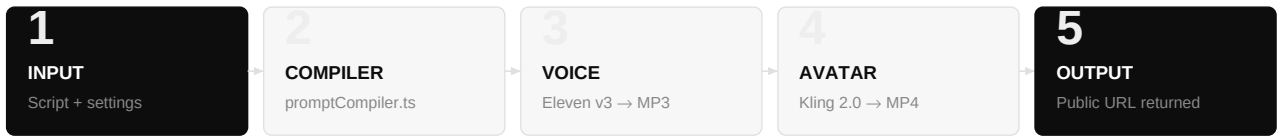
01 — SYSTEM OVERVIEW

ZAG is a prompt-driven avatar pipeline. Operator inputs script and selects modes. System compiles a structured prompt, generates voice via Eleven v3, and drives Kling Avatar 2.0.

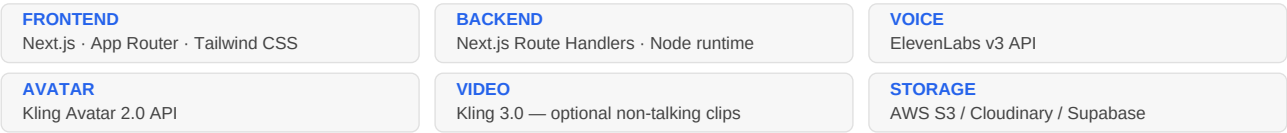
RESPONSIBILITY SPLIT



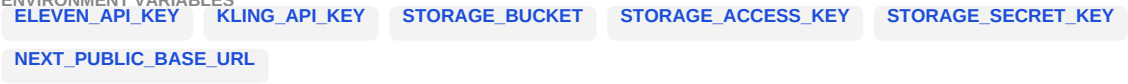
DATA FLOW PIPELINE



TECH STACK



ENVIRONMENT VARIABLES



02 — ARCHITECTURE & FOLDER STRUCTURE

PROJECT STRUCTURE

```
/app
page.tsx
/components
  PromptForm.tsx
  VideoPreview.tsx
/api/generate
  route.ts
/lib
  promptCompiler.ts
  eleven.ts
  klingAvatar.ts
  klingVideo.ts
/types
  index.ts
/public/assets
  zag_base.png
```

API ROUTE — /APP/API/GENERATE/ROUTE.TS

```
export async function POST(req) {
  const { script, promptType }
    = await req.json()

  // 1. Compile prompt
  const prompt = compilePrompt(
    script, promptType)

  // 2. Generate voice
  const audioUrl =
    await generateVoice(script)

  // 3. Generate avatar
  const videoUrl =
    await generateAvatar({
      image: "/assets/zag_base.png",
      audio: audioUrl,
      prompt,
      type: promptType
    })
}
```

MODULE RESPONSIBILITIES

<code>promptCompiler.ts</code>	Core logic. Assembles character, gesture, mouth, voice, and script blocks into a structured Kling prompt string.
<code>eleven.ts</code>	Calls ElevenLabs v3. Passes stability, similarity_boost, style. Uploads MP3 blob to storage. Returns public URL.
<code>klingAvatar.ts</code>	Sends image + audio URL + compiled prompt to Kling Avatar 2.0. Returns MP4 URL from animation response.
<code>klingVideo.ts</code>	Optional: Kling 3.0 non-talking clips. Used for B-roll, product shots, transition clips. Returns MP4 URL.

STORAGE OPTIONS

AWS S3 Production. Pre-signed TTL URLs. Scalable.	Cloudinary Easy setup. Built-in transforms. Dev-friendly.	Supabase Storage Open source. Good for MVP + auth integration.
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03 — APP WIREFRAME · FULL UI LAYOUT

Two-column layout · 380px sidebar · Flex preview panel · Dark theme on canvas area

ZAG

Avatar Generation · Must Be Nuts

SCRIPT

Type ZAG's script here...0 / 300

GESTURE MODE

Type A
Still + head pat

Type B
Head nod

CUSTOMIZE VARIANT

● Default

👤 Cost

👤 Out

👤 Cost

👤 Out

👤 Outfit

MOUTH ARTICULATION

Open Width8

Speed6

Jaw Close9

VOICE PRESET

SarcasticDeadpanHypeWhisperAggressive

VOICE TUNING

Stability7

Similarity6

Generate ZAG Video

Preview● Ready

Costume overlay

ZAG mascot · green screen

Status indicator

ZAG base image

Script ticker

GESTURE	OUTFIT	VOICE	EST.
Type A	Default	Sarcastic	~5s

compiled prompt ↓

Character: ZAG (crocodile mascot) · Visual Rules: Black suit, white gloves, Must Be Nuts mic...

04 — CONTROL PANEL · COMPONENT MAP

All interactive controls with output behaviour and compiled prompt impact

ID	COMPONENT	TYPE	OUTPUT / BEHAVIOUR	LIMIT	IMPACT
C1	Script Textarea	Text input	Injected as: Script: "<value>" — last block of compiled prompt	300 chars	HIGH
C2	Gesture Toggle	Binary A / B	Swaps gesture block — locked+head-pat (A) vs subtle motion (B)	A or B only	HIGH
C3	Costume Grid	6-option select	Replaces Visual Rules block with full costume description	6 options	HIGH
C4	Mouth Sliders	3x range 1–10	Appends [width:N/10], [speed:N/10], [close:N/10] to Mouth Mechanics block	1–10 each	MEDIUM
C5	Voice Preset	5-option chip	Sets style: field in Eleven v3 call + Preset: line in compiled prompt	5 presets	HIGH
C6	Voice Tuning	2x range 0–10	Maps to stability and similarity_boost in API body ElevenLabs (÷10)	0.0–1.0	MEDIUM
C7	Generate Button	Trigger CTA	POST /api/generate with all settings. Disabled is when script empty.	N/A	N/A

05 — PROMPT SYSTEM · VOICE CHANGES

5 voice presets — ElevenLabs API settings, delivery style, Kling sync instructions

SARCASTIC STAB SIM Slow, deliberate. Long pauses. Stress on unexpected words. Default ZAG delivery. "Oh wow. A nut. Incredible discovery." → Kling: Tone: Sarcastic, dry, controlled. Delivery: controlled, slightly slow.	stability: 0.7, similarity_boost: 0.6, style: "sarcastic"
DEADPAN STAB SIM Flat affect. Zero vocal variation. Reads like a government notice. No expression. "It is a snack. You eat it. Goodbye." → Kling: Tone: Flat, zero affect, robotic pace. No visible expression.	stability: 0.9, similarity_boost: 0.8, style: "deadpan"
HYPE STAB SIM Fast and punchy. Rising energy on every phrase. Maximum excitement. "THESE NUTS. ARE INSANE. EAT THEM NOW." → Kling: Tone: High energy, enthusiastic, fast. Punchy staccato cuts.	stability: 0.3, similarity_boost: 0.5, style: "energetic"
WHISPER STAB SIM Low, intimate. Close-mic feel. ASMR adjacent. Unexpectedly unsettling from a crocodile. "...you should probably eat these. just saying." → Kling: Tone: Hushed, intimate. Low volume. Minimal mouth movement.	stability: 0.8, similarity_boost: 0.7, style: "soft"
AGGRESSIVE STAB SIM Sharp, confrontational. Short sentences. Hard consonants. Zero warmth. "EAT. THE. NUTS. That is all." → Kling: Tone: Confrontational, sharp. Hard consonants. Zero warmth.	stability: 0.2, similarity_boost: 0.4, style: "aggressive"

06 — PROMPT SYSTEM · MOTION CHANGES

Gesture modes A / B and mouth mechanics slider-to-prompt mapping

GESTURE MODE A VS B — COMPILED PROMPT BLOCK

TYPE A — RESTRAINED

Gesture Rules:

- No major movement during speech
- Body locked, arms at sides
- At end: pats head in frustration
- Eyes: fixed forward gaze
- No neck tilt or lean

```
animation: {
  mouth: 'exaggerated',
  eyes: 'fixed',
  body: 'locked',
  gestures: 'A',
  end_action: 'head_pat'
}
```

TYPE B — EXPRESSIVE

Gesture Rules:

- Subtle hand gestures allowed
- Slight neck movement permitted
- Keep movement minimal
- No full arm raises or jumps
- Eyes: mostly fixed, blink OK

```
animation: {
  mouth: 'exaggerated',
  eyes: 'natural',
  body: 'minimal',
  gestures: 'B',
  neck_movement: 'slight'
}
```

MOUTH MECHANICS — SLIDER RANGE → PROMPT INJECTION

Each slider (1–10) appends a bracketed annotation to the Mouth Mechanics block in the compiled prompt.

SLIDER	RANGE	BEHAVIOUR	COMPILED PROMPT INJECTION
Open Width	1–3	Subtle lip part. Near-closed jaw.	Snout barely opens – minimal phoneme shaping
Open Width	4–7	Standard articulation. Natural croc jaw.	Snout opens mid-range – clear phoneme shaping
Open Width	8–10	EXAGGERATED. Max jaw drop per syllable.	Snout opens WIDE – cartoon-level mouth movement
Speed	1–3	Slow jaw. Drawl. Long vowels.	Mouth mechanics slow – deliberate open/close
Speed	4–7	Conversational pace. Natural delivery.	Mouth speed: natural conversational
Speed	8–10	Rapid fire. Staccato bursts.	Rapid mouth cycling – punchy staccato delivery
Jaw Close	1–3	Jaw stays loosely open between phrases.	Jaw does not fully close between phrases
Jaw Close	4–7	Partial close between phrases.	Jaw closes to resting position between phrases
Jaw Close	8–10	Snaps fully shut. Hard phrase gaps.	Jaw FULLY CLOSES between phrases – hard stops

07 — PROMPT SYSTEM · COSTUME / CLOTHES CHANGES

Each costume replaces the Visual Rules block. Base image (zag_base.png) stays constant.

DEFAULT

Black full-body mascot suit. White gloves. Must Be Nuts branded microphone.

Visual Rules:

- Black full-body mascot suit (velvet)
- White Mickey-style gloves
- Must Be Nuts branded microphone
- Fixed forward gaze, no eye movement

USE: Hero costume. All primary campaign content.

CHEF

White chef coat + toque. White gloves. Must Be Nuts mic. Recipe / snack content.

Visual Rules:

- White double-breasted chef coat
- Tall white chef toque (slightly crooked)
- White Mickey-style gloves
- Must Be Nuts branded microphone

USE: Recipe reels, snack content, food-adjacent campaigns.

SUIT

Black tailored blazer. White shirt. Thin black tie. White gloves.

Visual Rules:

- Black tailored blazer, single-button
- White dress shirt, thin black tie
- White Mickey-style gloves
- Must Be Nuts mic (kept for identity)

USE: Corporate parody, brand announcements, absurd talking head.

GYM

Red sleeveless gym tank. Black shorts. White sneakers. White gloves.

Visual Rules:

- Red sleeveless gym tank (fitted)
- Black athletic shorts, above-knee
- White athletic sneakers
- White Mickey-style gloves

USE: Fitness content, gym memes, workout fuel campaigns.

STREETWEAR

Oversized black graphic hoodie. Black joggers. White hi-tops. White gloves.

Visual Rules:

- Oversized black graphic hoodie
- Black slim jogger pants
- White hi-top sneakers
- White Mickey-style gloves

USE: Gen Z content, collab drops, streetwear crossovers.

FESTIVAL

Orange festival vest with badges. Sunglasses on snout. White gloves.

Visual Rules:

- Bright orange mesh festival vest
- Sunglasses resting on snout
- Badges and pins on vest
- White Mickey-style gloves

USE: Event content, music festivals, limited-run drops.

08 — FULL COMPILED PROMPT EXAMPLE

Type A · Chef Costume · Hype Voice · Mouth Width 9 · Speed 8 · Jaw Close 9

COMPILED PROMPT — sent to Kling Avatar 2.0

Character: ZAG (crocodile mascot)

Visual Rules:

- White double-breasted chef coat, slightly oversized
- Tall white chef toque (slightly crooked, comedic)
- White Mickey-style gloves
- Must Be Nuts branded microphone
- Black crocodile foam head unchanged

Mouth Mechanics:

- Snout opens wide during speech [width: 9/10]
- Fully closes between phrases [close force: 9/10]
- Exaggerated articulation [speed: 8/10]

Tone:

- Sarcastic, dry, controlled

Voice:

- Preset: Hype Stability: 0.3 Similarity boost: 0.5

Gesture Rules (Type A):

- No major movement during speech
- Body locked, arms at sides
- At end: pats head in frustration (single beat)

Performance:

- Direct to camera 9:16 vertical Green screen

Script:

"THIS. IS. THE BEST NUT YOU WILL EVER EAT.
I am a chef. I have made many nuts.
Stop questioning me. Just eat them."

ELEVEN V3 API CALL

```
POST https://api.elevenlabs.io/v1/generate

headers: {
  "xi-api-key": process.env.ELEVEN_API_KEY
}

body: {
  text: "[script]",
  voice_settings: {
    stability: 0.3,
    similarity_boost: 0.5,
    style: "energetic",
    use_speaker_boost: true
  }
}
```

KLING AVATAR 2.0 API CALL

```
POST https://api.kling.ai/avatar

body: {
  image: "/assets/zag_base.png",
  audio: audioUrl,
  prompt: compiledPrompt,
  animation: {
    mouth: "exaggerated",
    eyes: "fixed",
    body: "locked",
    gestures: "A",
    end_action: "head_pat"
  }
}
```

09 — PERFORMANCE · SCALING · ROADMAP

PERFORMANCE ARCHITECTURE

Async Job Queue	BullMQ	Avoids blocking Next.js route during Kling render (30–120s). Ret
Job Polling	GET /api/status/:jobId	Client polls every 2s. Returns { status: 'processing' 'done', vi
Storage Naming	zag_{jobId}_{type}.mp4	Prevents collision. Enables lifecycle cleanup. Pre-signed TTL UR
Rate Limiting	Per-user token bucket	Protects Kling + Eleven credits. 429 with retry-after header.
Error Handling	Step-level try/catch	Voice fail ≠ avatar fail. Granular error codes. Per-step retry I

FUTURE ROADMAP

VERSION	FEATURE	DESCRIPTION
V1.1	Batch Script Processing	Multiple scripts → job queue → batch. Status dashboard per job.
V1.2	Timeline Editor	Multi-clip stitching — avatar + B-roll + product shots → single export.
V1.3	Subtitle Auto-Gen	Whisper on audio → SRT → burned into final MP4.
V1.4	Gesture Intensity	Numeric 1–10 override for gesture intensity on top of A/B mode.
V1.5	Multi-Character	New mascots, same compiler. New character block, same pipeline.
V2.0	Campaign Mode	Full pipeline: batch + timeline + subtitles + brand overlay.

Next.js is NOT the intelligence layer.

It is the delivery and orchestration layer.

Intelligence lives in: prompt compiler · constraint enforcement · model chaining